54. Capstone: Broken Network Challenge

54. Capstone: Broken Network Challenge

Phase: Phase 13 — Capstones
Estimated time: 2–4 days
Slug: `54-capstone-debug`

Learning objectives

- Debug planted failures
- Produce timeline + root cause
- Fix with minimal change
- Write postmortem

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

Break a snapshot on purpose (wrong route, bad DNS, blocked SG, MTU, expired cert, hosts file typo). Restore service with postmortem:

Impact → Timeline → Root cause → Fix → Prevention

Common mistakes

- Recreate VMs instead of debugging
- Changing everything
- No postmortem

Exercises

Work in `exercises/54-capstone-debug/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/54-capstone-debug/` until after an honest attempt.

Self-check

1. Root cause?
2. Evidence?
3. Prevention?

Further reading

- https://sre.google/sre-book/postmortem-culture/